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Pattern of beech regeneration in the east-central United States.^{1,3}

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HELD, MICHAEL E. (Dept. Bot., Ohio Univ., Athens, Ohio 45701). Pattern of beech regeneration in the east-central United States. Bull. Torrey Bot. Club 110: 55-62. 1983.—*Fagus grandifolia* Ehrh. (American Beech) reproduces from seed and/or root sprouts. Although seedling must be the mechanism for initial establishment, root sprouting is considered by many researchers to be the main mode of reproduction in this species in certain areas of its range. Eight forested areas within the east-central portion of the range of *Fagus grandifolia* were selected for study of the environmental conditions related to the occurrence of root sprouts and/or seed-origin individuals. Multivariate statistical techniques such as principal component analysis, canonical correlation analysis and, stepwise regression analysis were used in this study to examine the relationship between environmental parameters and the reproductive activity in this species. Over the range of *Fagus grandifolia*, root sprouting is a mechanism by which this species survives in a forest within an area where environmental parameters are harsh. This study indicates a partial shift from seedling establishment toward a dependence on vegetative reproduction in areas where climate is more severe. Within each forest, however, the exposure of the slope on which a parent beech tree is located is an indicator of the possible origin of the beech juvenile individuals around that mature tree. These results do not preclude the occurrence of either root sprouts or of seed-origin individuals within any site in any area of the range of this species.

Key words: *Fagus grandifolia*; American beech; root sprouts; vegetative reproduction; deciduous forests; multivariate statistics.

Fagus grandifolia Ehrh. (American Beech) is one of several tree species which reproduce either by seed or vegetatively by the

production of root sprouts (Ward 1961; Bormann and Likens 1979). Little is known about the environmental conditions which affect the reproductive mechanisms in beech. The role of vegetative reproduction in its development and maintenance as a major component of many forest communities in eastern North America is not well understood (Forcier 1973, 1975; Keever 1973).

Although seedling must be the initial mechanism for establishment, root sprouting has been reported to be the main mode of reproduction in beech in certain areas of its range (Russell 1953; Rushmore 1961; Ward 1961; Haines 1965; Bormann *et al.* 1970; Forcier 1973, 1975; Keever 1973; Abrell and Jackson 1977). Seed mast in *Fagus* is sporadic and often the seeds contain non-viable embryos (Keever 1973). Although beech seeds are animal dispersed (Abrell and Jackson 1977), much of the viable mast is used as a food source for small forest mammals (Ward 1961).

The environmental conditions related to the mechanism of root sprouting in *Fagus grandifolia* have not been studied in detail over a large section of its range

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(Ward 1961). Curtis (1959) stated that, “The exact conditions responsible for such vegetative reproduction are not well understood and should be investigated further.” The role of root sprouting in hardwood trees as a successional mechanism has received attention recently (Forcier 1973, 1975; Held and Wistendahl 1977; Held 1978; Bormann and Likens 1979).

The objectives of the present study were 1) to describe the relationship between selected environmental variables and the seeding and sprouting activity of *Fagus grandifolia* within the central portion of the range of this species, and 2) to show the relationship of these reproductive mechanisms to geographic distribution of this species in the eastern deciduous forest of North America.

Study sites. Eight study sites within the central portion of the range of *Fagus grandifolia* in eastern North America were selected for study. Their topographic coordinates, geographic locations, and identification codes are presented in Table 1. These sites are more fully described in Held (1980).

Each site selected was either an old-growth forest or a second-growth stand not cut in 80 years preceding the sampling date. These criteria for site selection were used so that the environmental influences which affect forests would have had enough time to affect the development and maturation of *Fagus grandifolia* individuals in the forest. Each site was at least 10 ha in area, had a variety of slope inclinations and aspects,

and was located in the geologically and climatically diverse area of the east-central United States.

Materials and methods. Sampling at the Waterloo Wildlife Research Station was intentionally greater (100 plots) than at the other sites, in order to establish methods and sample size. Held and Wistendahl (1978), in a study at this site, recorded the tree-stratum composition around 52 variable-radius plots. The running means test (Kershaw 1973) and an estimate of the minimum number of variable-radius plots required to obtain a standard error within 5% of the mean showed that a sample of 14 variable-radius plots was sufficient to describe the community composition of upland forests at this site. A sample of 20 data point centers, approximately 50% over minimum (Kershaw 1973), was used to describe the tree composition and for data collection at each site. This sample size was used at the other seven sites.

For the eight sites, there was a total of 240 data points. Beech trees which served as data point centers were randomly selected, although each tree met the following criteria: at least 10 cm dbh and over 10 m in distance from the preceding tree selected. At each site, data points were obtained from all compass directions, if possible, and their specific locations were recorded as to position on slope. Data points were selected in a manner that would insure adequate representation of all available directions, thereby reducing

Table 1. List of eight study sites with county, state, topographic coordinates, meters above sea level, identification code for each site, and selected references.

Site	County	State	Topographic Coordinates	M Above Sea Level	Identification Code	Selected References
University of Wisconsin— Milwaukee Field Station Dysart Woods	Ozaukee Belmont	WI OH	(88°01'W 43°23'N) (80°59'W 39°58'N)	270 381	OCWI BCOH	Dunnum, 1972a,b Levenson, 1976 Lafer and Wistendahl, 1970 Nixon, 1963 Nixon, <i>et al.</i> , 1968
Waterloo Wildlife Research Station	Athens	OH	(82°15'W 39°20'N)	238	ACOH	
Boone County Cliffs Nature Preserve	Boone	KY	(84°47'W 39°00'N)	230	BCKY	Bryant, 1978 Cain, 1932
Donaldson's Woods Nature Preserve	Lawrence	IN	(86°24'W 38°44'N)	200	LCIN	Lindsey <i>et al.</i> , 1958
Myers' Beech Woods	Monroe	KY	(85°49'W 36°44'N)	275	MCKY	
Tech Aqua Field Station	Dekalb	TN	(85°44'W 36°02'N)	236	DCTN	
Walker Cove Research Natural Area	Buncombe	NC	(82°21'W 35°45'N)	1275	BCNC	

sampling bias. Data were obtained from a circular plot with a 5 m radius (0.008 ha) surrounding the selected tree. Within each circular plot all juvenile beech stems were counted. The developmental pattern (root sprout or seed-origin) of each beech stem was determined by excavating the root system of each individual plant to a point at which a determination of the origin of the stem as sprout or seed could be made.

The community composition in the vicinity of each circular plot was determined using the variable-radius method of plotless forest sampling (Grosenbaugh 1952), employing a calibrated glass prism. The central tree in each plot served as the center point from which community composition data were collected.

A list of the variables measured at each data point is presented in Table 2. Three of the 14 variables (GROUP, EXPO, and POS) were non-continuous and were represented by categorical or "dummy" variables (Finn 1974). This expansion of the three non-continuous variables increased the number of variables to 28, and these were included in the statistical analysis.

A soil sample of 100g was collected within a circular band that was 1 m wide and at least 1 m from the central tree in the plot. The soil was collected from the upper 10 cm level excluding the humus and litter. All soil samples, less one sample because one tree was growing on a rock outcrop, were returned to the laboratory, air dried,

and passed through a 2 mm sieve in preparation for physical and chemical tests.

Soil factors which were measured using the soil samples included: soil texture, soil pH, and the concentrations (in ppm) of calcium, potassium, and magnesium. Soil texture was determined by the hydrometer method (Bouyoucos 1951). Soil pH was determined according to the method of Peech (1965). Determinations of the concentrations (ppm) of calcium, potassium, and magnesium in the soil samples were made using a Perkin-Elmer Model 360 Spectrophotometer. Exchangeable cation concentration levels were determined using the ammonium acetate extraction technique, modified from Chapman (1965).

Topographic data obtained at each point included the percent slope angle, the aspect, and the position of the central tree on the slope. Stand structure variables that were measured included basal area of the central beech tree in each plot, density of root sprouts, and seed-origin stems within each plot.

In all the analyses, the variables, "the number of root sprouts" (SPROUT) and "the number of seed-origin individuals" (SEED), within a data plot, were considered to be dependent variables. All the remaining variables (26 measures) were considered to be independent or predictor variables. The data matrix was of the size 240×28 , with 28 variables and 240 observations. This data matrix was used in all

Table 2. List of variables measured at each data point. Codewords are given with an explanation of each variable.

SPROUT—number of root sprouts
SEED—number of seed-origin individuals
BASAL—basal area of central tree in each data point
SLOPE—percent slope
SAND—percent sand in soil sample
CLAY—percent clay in soil sample
SILT—percent silt in soil sample
CA—ppm calcium in soil sample
K—ppm potassium in soil sample
MG—ppm magnesium in soil sample
PH—pH of soil sample
GROUP ^a —geographical location of study site
EXPO ^b —exposure of slope where data point is located
POS ^c —position of data point on slope

^a This variable was represented by eight categorical variables for data analysis.

^b This variable was represented by seven categorical variables for data analysis.

^c This variable was represented by two categorical variables for data analysis.

Table 3. The number, exposure occurrence, and % occurrence of root sprouts (RS) and seed-origin individuals (SO) at each site. (Site codes identification given in Table 1.)

Density and occurrence of root sprouts and seed-origin plants	Study Site															
	OCWI		BCOH		ACOH		BCKY		LCIN		MCKY		DCTN		BCNC	
	RS	SO	RS	SO	RS	SO	RS	SO	RS	SO	RS	SO	RS	SO	RS	SO
NE	42	7	206	33	15	1446			1	17		6	1	70	33	245
NW	59	25	22		5	520	3	33		7				6	7	33
S	29				2	226							1		15	35
SE	84	22	10	33	16	154		1		26	1	8		4	5	22
SW	16	6	110	217	50	520	1	13		2	5	69				
W	6	2			6	30		6		13		27				
Total	236	62	326	283	94	2488	4	53	1	65	6	110	2	162	60	335
% occurrence	95	95	75	100	36	95	10	70	5	90	15	85	10	100	8	100

statistical analyses. All statistical computations were made on the IBM 370/45 at either the Ohio University Computer Center or The University of Tennessee, Knoxville, Computer Center using selected statistical programs from the Biomedical Computer Programs (Dixon 1975), Statistical Analysis System (Barr *et al.* 1976), and the general multivariate statistical analysis program, MULTIVAR (Finn 1974, 1976).

Results. FIELD STUDIES. The pattern of root sprout occurrence was observed to increase from the central and southern portion of the study area to the northern sites. Also, there was a positive relationship between increasing sprout density and elevation in the southern sites. These patterns are also apparent in the occurrence frequency of sprouts. There is no interpreta-

ble pattern between seedling density and frequency and the various study sites (Table 3).

Soil data indicated a wide range of variation for the variables measured—hydrogen ion concentration, for example—across the 240 samples at the eight study sites. Ca, K, and Mg also showed great variability (Table 4). This variation may be an artifact caused by the method of sampling. Sites were visited at different times of the year, and thus samples were collected at different periods of the geochemical cycle of the soil. Likens *et al.* (1977) observed seasonal variation in the budget of various nutrients at the Hubbard Brook forest. Applicable methods of weighting data to eliminate this internal variation have not been successful in this study. Soil textures, while not varying seasonally, showed variation within and be-

Table 4. Soil texture class frequency and ranges of pH, calcium, potassium, and magnesium levels at each site. (Site codes identification given in Table 1.)

Soil sample-class density	Study Site							
	OCWI	BCOH	ACOH	BCKY	LCIN	MCKY	DCTN	BCNC
loam	5	8	36	6		7	16	2
sandy loam	12		27	6			3	8
sandy clay loam	3		20	1			1	10
silt loam		7	7	6	12	13		
silty clay loam		2			6			
clay loam		2	8		2			
clay		1		1				
sandy clay			1					
pH range	5.2-7.4	5.3-6.4	4.8-6.6	5.3-7.6	5.6-7.1	5.9-6.7	5.2-6.3	5.2-6.2
Ca (ppm) range	28-142	2.0-46	1.0-47	3.0-73	8-60	5-47	4-40	8-68
K (ppm) range	30-145	1.6-6.8	0.6-3.0	1.4-9.5	50-140	3.4-9.5	65-195	105-270
Mg (ppm) range	60-332	0.4-8.6	0.2-11.2	0.6-5.5	24-72	1.0-3.0	20-148	32-136

tween sites and thus were difficult to interpret (Table 4).

The other variables selected for study, the community and topographic measures, are stable throughout the year, and sample variability is not time-dependent. When comparing slope aspect to the occurrence of sprouting and seed-origin development over the eight sites, a pattern emerges. At five of the sites, the majority of root sprouts at a given site were found on west-facing slopes; the same pattern was seen for seed-origin stems except that they were concentrated on east facing slopes (Table 3).

STATISTICAL ANALYSIS. Principal component analysis was used to reduce the number of independent variables that would be used in further analyses (Noy-Meir 1970). This analysis (not shown) resulted in 26 components that were extracted from the data; however, only 10 had eigenvalues greater than 1.00 and were considered to be significant. Only the first four components were interpretable, that is, had component loadings of greater than ± 0.50 . The first and second components appeared to be the soil texture and soil chemistry axes. These components accounted for approximately 32% of the variation in the sample. As discussed previously, the variability in the soil chemistry data may be an artifact of the sampling procedure, and these variables were not included in further analyses. The third and fourth components were characterized as exposure axes. On both components there were high loadings for some of the exposure measures, and over both components all the exposure variables had high (greater than ± 0.50) loadings. These two components accounted for approximately 15% of the total variation.

Canonical correlation analysis was performed on the combined data collected at all study sites. In the analysis, the dependent variables were SPROUT and SEED, and the independent variables were considered to be the exposure variables. In canonical correlation analysis, the number of canonical correlations that can be extracted is equal to the number of dependent variables. However, in actual analysis, one or both canonical correlations may not be significant and are thus considered non-interpret-

able (Anderson 1958). In the analysis of the combined data, the second canonical correlation was insignificant; it was deleted from the analysis. The results of the first canonical correlation (not shown) indicated a positive relationship between seed-origin stems and northeast-facing slopes and a negative relationship with southwest-facing slopes. This analysis suggested that seed-origin stems occurred predominantly on northeast rather than southwest aspects.

Because canonical correlation analysis did not provide any interpretable results concerning the relationship of sprouting occurrence and the exposure variables, backward stepwise regression analysis was used. The backward elimination method was selected because, in this procedure, the first equation that is derived contains all the independent variables and then produces equations with variables being eliminated until certain stopping criteria are met (Chatterjee and Price 1977). The equation derived with sprouting as the dependent variable was $\text{SPROUT} = 0.06 + 0.43 \text{ NW} + 0.15 \text{ SE}$, indicating that northwest and southeast exposures were useful in determining sprout occurrence.

Discussion. Although sexual reproduction is a common regenerative mechanism in *Fagus grandifolia* (Ward 1961; Forcier 1973, 1975), root sprouting was also observed at the eight study sites. The occurrence or non-occurrence of root sprouts is indicative of the geographic area in which beech individuals are found and also of microhabitat conditions within each forest.

The results of this work indicate that at the study sites there is a positive association between the occurrence of seed-origin individuals and northeast aspects and a negative relationship with southwest aspects. For root sprout individuals, west and south facing slopes appear to be important predictors in characterizing the occurrence of these asexually produced individuals. These results indicate that within the study sites, although beech individuals regardless of origin occur on all exposures, there is a segregation of root sprouts and seed-origin individuals on separate aspects.

Cantlon (1953), in his study of microcli-

mate on different slopes on Cushetunk Mountain, New Jersey, found that on south slopes, the ambient and soil temperatures were consistently higher than the temperatures on north exposures. This implies that north aspects retain moisture in the soil; whereas, south exposures are drier and more susceptible to erosion. Throughout the present study, the majority of the root sprouts were recorded as occurring on shallow or exposed roots; whereas, seed-origin individuals occurred on more stable soils. Ward (1961) reports that, in Wisconsin, the root sprouts he observed were, with minor exception, found on lateral roots near or at the surface. He states that abrasion of the roots may stimulate the development of root sprouts. Cantlon (1953) states that there is an earlier start of activity on south aspects. With this earlier beginning of activity, there is a possibility of freeze-thaw damage to the roots. Sun scald injuries to exposed roots may initiate root sprout development. The early warming on south slopes may promote beech seed germination; thus, any late frost may kill the developing seedlings and eliminate many of them from south exposures. On north slopes, the opposite conditions are often present.

The exposure variables on an individual forest within the east-central portion of the range of *Fagus grandifolia* can be used to predict the occurrence and abundance of root sprout and seed-origin individuals. However, neither root sprouts nor seedlings are limited to any slope aspect. The slope exposure can only give an indication to the possible environmental parameters which may influence the type of development in a particular stand.

The abundance of root sprouts is also a function of geographic space (macrohabitat parameters); whereas, the occurrence of root sprouts is a function of microhabitat parameters. Canonical correlation analysis and stepwise regression analysis were used to analyse these microhabitat data.

A segregation of sites based upon predominance of individuals originating from seed or root could be observed (Table 3). At sites in the central and western regions of the range of *Fagus grandifolia* (Myers' Beech Woods, Boone County Cliffs Nature

Preserve, Tech Aqua Field Station, and Donaldson's Woods Nature Preserve) few root sprouts were recorded. Ward (1961) found this pattern of few root sprouts in forests in Indiana and western Tennessee. However, at the other four sites, many root sprouts were observed. Two of these sites (Walker Cove Research Natural Area and The University of Wisconsin, Milwaukee Field Station) are located in extreme portions of the range of sites studied and have more harsh environments in comparison to the other six sites which are located in the central section of the range. In these environments, there is a greater possibility of freeze-thaw activity in the early spring, and also, the dieback of young seedlings due to late spring frosts. At the other sites, the date of the last frost occurs early in the spring, minimizing the possibility of frost damage. Russell (1953) reports that, in the high elevation "beech gaps" of the Great Smoky Mountains National Park, there were no juvenile beech individuals of seed-origin to be found. He states that in these "beech gap" areas of the mountains, there are greater extremes of temperature than at lower elevations. However, beech seedlings have been reported in these high elevation gaps (Sain and Blum 1981). Ward (1961) reports that there were very few individuals of seed-origin in southeast Wisconsin and that the main mode of reproduction is by root sprouting. He also states that, although there are more seedlings in northeast Wisconsin forests, there is a general increase in root sprouts from southern to northern Wisconsin.

Sprouting from roots in *Fagus grandifolia* is of importance to its survival and maintenance within the forests of eastern North America. This species, once removed from a community, is lost to that community, because of its production of mast at irregular intervals and poor mobility of its large, heavy seeds (Forcier 1973). Sprouting assures the continued production of stems should seed production be low and predation of seed by small mammals of the forest be high (Schroger 1949). An interesting question for future work deals with the frequency and occurrence of non-viable embryos (Keever 1973) and root

sprouting in this species. The production of root sprouts may also be a process by which beech maintains its dominance in the community and delays the invasion of other tree species and leads to an increased importance of beech in the community (Forcier 1973, 1975).

Root sprouting is possibly a mechanism by which the species survives locally within the range where climatic and other habitat parameters are severe. The results of this study imply a partial shift from successful seedling establishment toward a dependence on vegetative reproduction in those areas of its range (high elevations or northern latitudes) where the environment is more severe. Within each forest site, microhabitat forces influence the relative occurrence and abundance of root sprouts and seed-origin plants. The aspect of the slope on which a parent tree occurs is an indicator of the possible origin of juvenile beech plants around mature trees. These conclusions do not preclude the occurrence of either root sprouts or of seed-origin individuals within any site in the range of *Fagus grandifolia* or on any exposure within a forest stand examined.

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